

1. Administrative & Study Identification

Full Study Title: A Two-Part, Phase 1a/b, Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF-07284890 (ARRY-461) in Participants With BRAF V600 Mutant Solid Tumors With and Without Brain Involvement **Short Title/Acronym:** A FIH Study of PF-07284890 in Participants With BRAF V600 Mutant Solid Tumors With and Without Brain Involvement **Protocol Number:** C4471001 **NCT Number:** NCT04543188 **Sponsor Name:** Pfizer **Investigational Product:** PF-07284890 (ARRY-461) as monotherapy and in combination with Binimetinib **Phase of Development:** Phase I (Phase 1a/1b) **Medical Monitor:** Pfizer Clinical Operations Lead / Principal Investigator

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: * *Phase 1a:* To determine the Cycle 1 dose-limiting toxicities (DLTs), Maximum Tolerated Dose (MTD), and/or Recommended Dose for Expansion (RDE). Additionally, to evaluate the number of participants with Treatment-Emergent Adverse Events (TEAEs) and hematology laboratory abnormalities. * *Phase 1b:* To evaluate the overall response (ORR) per RECIST v1.1 (overall and intracranial) and RANO criteria for primary brain tumors. **Secondary Objectives:** To assess the duration of response (DOR), progression-free survival (PFS), disease control rate (DCR), time to response (TTR), overall survival (OS), adverse events (AEs), and specific pharmacokinetic (PK) parameters (including C_{max}, T_{max}, and AUC_{last}).

2.2 Background & Rationale BRAF inhibitors have transformed the treatment for patients with BRAF V600-mutant cancers (such as malignant melanoma, non-small-cell lung carcinoma, primary brain neoplasms, and other malignant neoplasms). However, their long-term efficacy is significantly limited by disease progression in the central nervous system (CNS) due to poor blood-brain barrier (BBB) penetration. PF-07284890 is a highly potent, selective, and brain-penetrant small-molecule inhibitor of BRAF V600 mutations designed to cross the BBB and achieve therapeutic concentrations in the brain, potentially improving outcomes for patients with brain metastases and primary brain tumors.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Single-arm (Dose-escalation/expansion) **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target **\$N\$:** 65 participants (Actual final enrollment prior to termination) **Number of Sites:** Multicenter (e.g., Emory University, Johns Hopkins University, Dana-Farber Cancer Institute) **Estimated Study Start:** 08-Jan-2021 (Actual) **Estimated Primary Completion:** 20-Mar-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 16 years at the time of consent. 2. Histologically confirmed advanced/metastatic solid tumor (including primary brain tumor), documented evidence of BRAF V600 mutation in tumor tissue or blood, with or without brain involvement. Must have adequate tumor tissue available for central laboratory submission. 3. Disease progression despite prior treatment, with no acceptable alternative treatment options available. Prior treatment rules apply to Phase 1b Dose Expansion: * *Cohorts 1 & 2:* No prior BRAF inhibitor in the metastatic setting or adjuvant setting within 6 months. * *Cohorts 3 & 4:* Required prior BRAF inhibitor in the metastatic or adjuvant setting within 6 months. 4. Specific Phase 1b Dose Expansion Cohort Requirements: * *Cohorts 1, 2, 3, 4:* Melanoma with at least 1 parenchymal brain lesion (Cohorts 1 & 3 asymptomatic for ≥ 14 days; Cohorts 2 & 4 symptomatic within 14 days prior to start). * *Cohort 5:* Any solid tumor not meeting Cohorts 1-4 criteria; history of or current leptomeningeal metastases allowed. * *Optional Cohort 6 (DDI Sub-study) & Cohort 7 (Food-Effect):* If brain involvement is present, participants must be asymptomatic. 5. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1.

4.2 Exclusion Criteria 1. Brain metastasis or primary brain tumor requiring immediate local intervention. 2. History of or current leptomeningeal metastases (unless enrolling in specific Cohort 5). 3. Any other active malignancy within 2 years prior to enrollment. 4. Prior radiation therapy to visceral metastases within 14 days, or whole-brain radiation therapy (WBRT) within 28 days prior to study treatment. 5. Systemic anti-cancer therapy or small-molecule therapeutics within 2 weeks prior to start of study treatment; antibody-based agents within 4 weeks prior to start. 6. History or current evidence of Retinal Vein Occlusion (RVO) or current risk factors for RVO; history of retinal degenerative disease.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Starting dose of PF-07284890 is 50 mg QD. When used in combination, binimetinib is dosed at 45 mg BID. For the Optional Cohort 6 (DDI Sub-study), midazolam is administered to evaluate drug-drug interactions.

Route of Administration: Oral **Duration of Treatment:** Treatment continued until disease progression, unacceptable toxicity, or patient refusal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for advanced malignancies. **Prohibited:** Systemic anti-cancer therapy or small-molecule therapeutics within 2 weeks prior to start of study treatment; antibody-based agents within 4 weeks prior to start.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Molecular Testing (BRAF V600), Baseline Scans (CT/MRI)
Phase 1a (Dose Escalation)	Cycle 1 (Days 1-28)	PK sampling, DLT monitoring, First Dose, Safety Labs
Phase 1b (Dose Expansion)	Cycle 2+	Efficacy Assessment (RECIST/RANO), Safety Follow-up
End of Study	+30 Days post-dose	Final Vital Signs, Exit Interview, IP Accountability, End of Treatment Scans

7. Outcome Measures (Endpoints)

7.1 Primary Endpoints * **DLTs:** Number of participants with Dose-Limiting Toxicities (DLTs) in Cycle 1 (first 21 days) to determine MTD/RDE (Phase 1a). Graded per NCI-CTCAE v5.0. *

Adverse Events: Number of participants with Treatment-Emergent Adverse Events (TEAEs), Serious TEAEs, Serious Treatment Related TEAEs, and Grade 3, 4, or 5 TEAEs evaluated by NCI-CTCAE v5.0. *

Laboratory Safety: Number of participants with Hematology Laboratory Abnormalities of Any CTCAE Grade (Phase 1a). *

Efficacy: Overall Response Rate (ORR) (Phase 1b). Measured using clinical and central radiologic tumor assessments via RECIST 1.1 or RANO criteria.

7.2 Secondary & Exploratory Endpoints * Duration of response (DOR), Progression-free survival (PFS), Disease control rate (DCR), and Overall survival (OS). * Plasma pharmacokinetic (PK) parameters for single dose PF-07284890 and Binimetinib (Phase 1a): * Maximum Observed Concentration (C_{max}) * Time for C_{max} (T_{max}) * Area Under the Plasma Concentration Time Curve From Time Zero to the Last Time Point of Quantifiable Concentration (AUC_{last})

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of TEAEs (defined as events starting from the first dose up to 30 days post-last dose or start of new therapy), clinical hematology and laboratory abnormalities, and tracking of dose interruptions/reductions. All AEs graded per NCI-CTCAE v5.0. **Serious Adverse Events (SAEs):** Reported per standard protocol regulatory requirements (suspected to cause death, life-threatening, requiring hospitalization, causing persistent disability/incapacity, or congenital anomalies). **Data Monitoring Committee (DMC):** Safety monitoring and dose-level decisions informed by an adaptive Bayesian Logistic Regression Model during the dose-escalation phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for efficacy, Safety Set for all participants receiving ≥ 1 dose of the investigational product. **Sample Size Justification:** Approximately 35 patients were originally targeted for Phase 1a to determine MTD/RDE via Bayesian Logic Regression Model. Final targeted enrollment across all study arms reached 65 participants. **Handling of Missing Data:** Participants with missing efficacy data were considered non-responders for ORR. Data rules aligned with standard oncology phase 1 trial imputation frameworks.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study was conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. *Note: The Sponsor (Pfizer) terminated the study due to an internal business decision to discontinue further development of PF-07284890. The termination was not due to major safety concerns or requests from regulatory authorities.* **Monitoring Plan:** Routine clinical monitoring for source data verification, safety reviews, and proper RECIST/RANO application. **Audit Trail:** EDC systems utilized for eCRF locking and query management.

11. Study Identifiers & Governance

Primary ID: C4471001 **Registry Number:** NCT04543188 **Sponsor/Lead Organization:** Pfizer **Study Title:** A FIH Study of PF-07284890 in Participants With BRAF V600 Mutant Solid Tumors With and Without Brain Involvement **Official Regulatory Title:** A Two-Part, Phase 1a/b Open-Label, Multicenter Trial Evaluating Pharmacokinetics, Safety and Efficacy of PF-07284890 (ARRY-461) in Participants with BRAF v600-Mutant Solid Tumors With and Without Brain Involvement

12. Clinical Framework (Oncology Specific)

Primary Condition: BRAF V600-Mutant Solid Tumors (including Malignant Melanoma, Non-Small-Cell Lung Carcinoma, Primary Brain Neoplasms, and other Malignant Neoplasms, with or without brain involvement) **Diagnosis Threshold:** Histologically confirmed mutation via standardized genomic sequencing in tissue or blood **Medical Methodology:** Targeted Kinase Inhibition (BRAF and MEK inhibitors) **Optimization Target:** Maximum Tolerated Dose (MTD) and Recommended Dose for Expansion (RDE)

13. Study Procedures & Methodology

Management Pathway: Stepwise dose-escalation followed by cohort-specific dose-expansion (including targeted melanoma cohorts, specific brain metastasis populations, leptomeningeal cohorts, and DDI/food-effect sub-studies) **Education Protocol (HCP):** Protocol training, AE management, and CTCAE v5.0 grading calibration **Education Protocol (Patient):** Informed consent process, oral medication adherence training, and AE reporting logs **Quality Audit Mechanism:** Central imaging reviews (RECIST v1.1 and RANO criteria) to ensure standardized efficacy reads

14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Cycle-based monitoring (e.g., Days 1 and 15 of early cycles for PK/safety) **Remote Monitoring Frequency:** As permitted by institutional standards during the trial duration **Checklist Review Interval:** End of each treatment cycle (typically every 21 to 28 days) prior to dispensing the next cycle's IP

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				